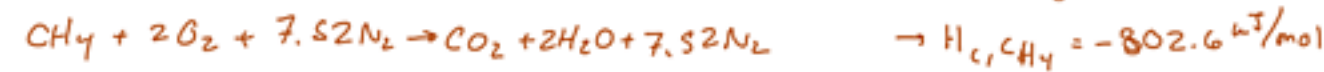
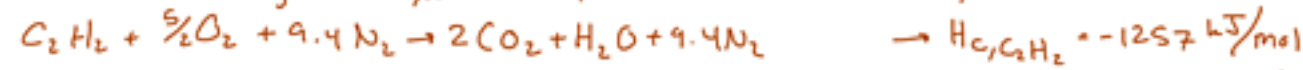


Q1. Murphy 6.47

Fuel mixed w/ air @ 25°C & reacts adiabatically....

Acetylene is used in welding torches, but more expensive than methane. Compare....



Adiabatic....

$$C_{p,out} = \sum_i \gamma_i C_{p,i}$$

$$C_{p,CO_2} = 37$$

$$C_{p,N_2} = 33.6$$

$$C_{p,H_2O} = 29.1$$

basis: 1 mol

$$1257 = \int_{298.15}^{T_2} C_{p,out} dT \quad |^2|, H_{C_2H_2} = -1257 \text{ kJ/mol}$$

$$1257 = (T_2 - 298.15) \left[(2 \times 37) + (33.6) + (9.4 \times 29.1) \right] \left(\frac{1}{1000} \right)$$

$$1257 = (T_2 - 298.15)(0.38114)$$

$$3298 = (T_2 - 298.15)$$

$$3596.15 \text{ K} = T_2$$

$$802.6 = \int_{298.15}^{T_2} C_{p,out} dT \quad |^2|, H_{CH_4} = -802.6 \text{ kJ/mol}$$

$$802.6 = (T_2 - 298.15) \left[(1 \times 37) + (2 \times 33.6) + (7.52 \times 29.1) \right] \left(\frac{1}{1000} \right)$$

$$= 0.323(T_2 - 298.15)$$

$$2484.8 = T_2 - 298.15$$

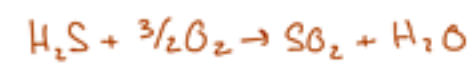
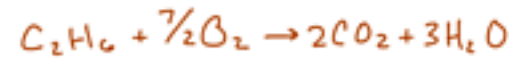
$$2782.95 = T_2$$

w/ the calculated outcomes of their respective temperatures, it is shown why one is more expensive, as it can get a higher temperature than the other.
(acetylene) (methane)

Q2. Murphy 6.48

A waste gas stream contains 40 mol% ethane (C_2H_6), 19 mol% N_2 , 1 mol% H_2S , 10 mol% O_2 , & 30 mol% CO . Fed into furnace w/ 20% excess air & combust. 25°C exits 150°C

... Complete combustion...



$$H_{C_2H_6} = -1428.6 \text{ kJ/mol}$$

$$H_{H_2S} = -518.7 \text{ kJ/mol}$$

basis: 1 g mol

$$Q_w = \dot{H}_{C_2H_6} + \dot{H}_{H_2S} + \int_{298.15}^{423.15} \eta_f C_{p,i} dT = \sum_c \eta_c \hat{H}_c - \sum_f \eta_f \hat{H}_f \quad \left| \begin{array}{l} \dot{H}_1 = 0.7 \text{ g/mol} \\ \dot{H}_2 = 0.01 \text{ g/mol} \end{array} \right.$$

Species	moles in	$\hat{H}_{inlet}$ @ 25°C	moles out	$\hat{H}_{exit}$ @ 150°C
C_2H_6	0.4	(-83.82)	0	—
N_2	6.388	0	6.388	(3.652)
H_2S	0.01	(-19.96)	0	—
O_2	1.698	0	0.283	3.736
CO	0.3	(-110.53)	0.3	(-112.083)
H_2O	0	—	1.2	(-237.562)
SO_2	0	—	0.01	(-291.543)
CO_2	0	—	0.3	(-388.531)
b.b.		$= -66.8866 \text{ kJ}$		$= -608.053 \text{ kJ}$

$$\left(\frac{7}{2} \times 0.4 \right) + \left(\frac{3}{2} \times 0.01 \right) (1.2) = 1.698 \text{ moles of } O_2$$

$$1.698 / 0.21 = 0.799 \text{ moles of } N_2$$

$$\eta_{O_2} = 1.698 - (3.5 \times 0.4) - (1.5 \times 0.01) = 0.283$$

$$\text{Nitrogen: } \int_{298.15}^{423.15} \left[31.15 - (1.357 \times 10^{-2})T + (2.68 \times 10^{-5})T^2 - (1.48 \times 10^{-8})T^3 \right] dT = 3.65 \text{ kJ/gmol}$$

$$\text{oxygen: } \int_{298.15}^{423.15} \left[23.11 - (2.68 \times 10^{-4})T + (1.746 \times 10^{-6})T^2 - (1.665 \times 10^{-9})T^3 \right] dT = 3.73598 \text{ kJ/gmol}$$

$$CO: -110.53 + \int_{298.15}^{423.15} \left[30.87 - (0.129T) + (2.789 \times 10^{-3})T^2 - (1.272 \times 10^{-6})T^3 \right] dT = -112.083 \text{ kJ/gmol}$$

$$CO_2: -393.5 + \int_{298.15}^{423.15} \left[19.8 + (0.07344T) - (5.602 \times 10^{-5})T^2 + (1.7115 \times 10^{-8})T^3 \right] dT = -388.531 \text{ kJ/gmol}$$

$$SO_2: -296.81 + \int_{298.15}^{423.15} \left[23.85 + (6.6699T) - (4.961 \times 10^{-5})T^2 + (1.328 \times 10^{-8})T^3 \right] dT = -291.543 \text{ kJ/gmol}$$

$$H_2O: -241.83 + \int_{298.15}^{423.15} \left[32.24 + (0.001924T) + (1.055 \times 10^{-5})T^2 - (2.596 \times 10^{-9})T^3 \right] dT = -237.562 \text{ kJ/gmol}$$

Plug in to the heat balance...

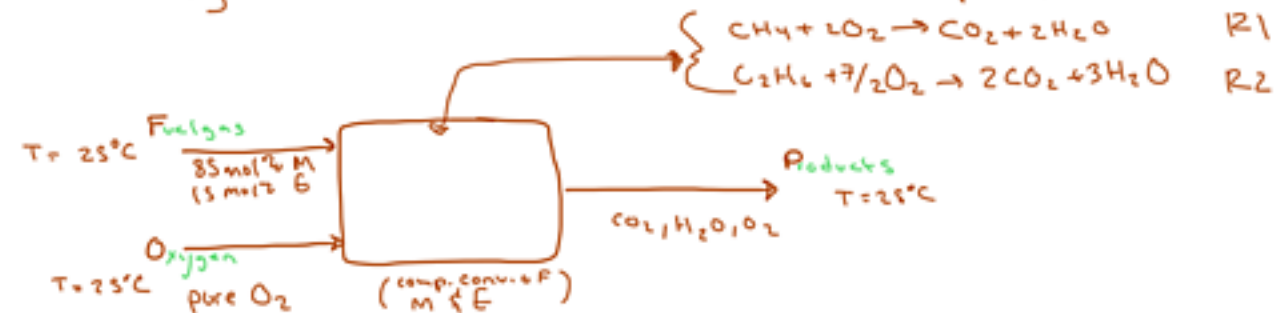
$$= (-608.053) - (-66.8866) = -541.166$$

Heat from waste = heat used to change the stream $\rightarrow -Q_w = Q_s$

$$Q_s = m \Delta H = m(50069.07 - 3785.06)$$

$$\frac{541.166 \text{ kJ}}{46.28 \text{ g}} = m = 0.011692 \text{ kmol} \times \frac{1000 \text{ mol}}{1 \text{ kmol}} \times 12.015 \text{ g/mol} = 210.6371 \text{ g} \approx 211 \text{ g}$$

Q3. A fuel containing 85 mol % methane, ethane, is burned completely +



a) Suppose the reactor is continuous... take basis of 1 mol/s of fuel gas + combust... ..

$$\dot{Q} = \dot{Q}/\dot{n} \text{ in kJ/mol...}$$

Basis: 1 mol/sec of fuel gas

$$\text{O}_2 \text{ needed for R1: } (0.85 \frac{\text{mol}}{\text{s}}) \times \frac{2 \text{ mol O}_2}{1 \text{ mol CH}_4} = 1.7 \text{ mol/s O}_2$$

$$\text{... for R2: } (0.15 \frac{\text{mol}}{\text{s}}) \times \frac{7/2 \text{ mol O}_2}{1 \text{ mol C}_2\text{H}_6} = 0.525 \text{ mol/s O}_2$$

$$\text{but, O}_2 \text{ supplied to furnace} = (1.7 \text{ mol/s O}_2) + (0.525 \text{ mol/s O}_2) = 2.225 \text{ mol/s O}_2$$

$$\text{Heat gen. in furn. } \dot{Q} = \dot{n}_{\text{fuel}} \Delta \hat{H}_r + \sum \dot{n}_i \hat{H}_i$$

left entered @ same temp.

$$\dot{Q} = \sum \dot{n}_i \Delta \hat{H}_r$$

$$\dot{Q} = [10.85 \text{ mol}(-802.6) + (0.15 \text{ mol})(-1428.6)] \text{ kJ}$$

$$= -896.5 \text{ kJ}$$

$$\dot{Q} = -896.5 \text{ kJ/s}$$

$$\dot{Q} \rightarrow -896.5 \text{ kJ/mol}$$

b) rework the problem for a constant-vol. batch reactor... Use basis of 1 mol of fuel...

$$\text{O}_2 \text{ for R1: } (0.85 \text{ mol}) \times \frac{2 \text{ mol O}_2}{1 \text{ mol CH}_4} = 1.7 \text{ mol O}_2$$

$$\text{for R2: } (0.15 \text{ mol}) \times \frac{7/2 \text{ mol O}_2}{1 \text{ mol C}_2\text{H}_6} = 0.525 \text{ mol O}_2$$

$$\text{but } = 2.225 \text{ mol O}_2$$

$$\dot{Q} = \sum n_i \Delta \hat{H}_i + \hat{U}_{\text{H}_2\text{O}}$$

$$\Delta \hat{H}_{\text{H}_2\text{O}} - RT = 46.65 \text{ kJ/mol} - (8.314 \text{ J/mol}\cdot\text{K})(298 \text{ K})(\frac{1}{1000})$$

$$\hat{U}_{\text{H}_2\text{O}} = 38.17 \text{ kJ/mol}$$

$$\dot{Q} = [n_{\text{CH}_4} \Delta \hat{H}_{\text{CH}_4} + n_{\text{C}_2\text{H}_6} \Delta \hat{H}_{\text{C}_2\text{H}_6}] + n_{\text{H}_2\text{O}} \hat{U}_{\text{H}_2\text{O}}$$

$$= [(0.85 \text{ mol})(-802.6 \text{ kJ/mol}) + (0.15 \text{ mol})(-1428.6 \text{ kJ/mol})] + [(2.225 \text{ mol})(38.17 \text{ kJ/mol})]$$

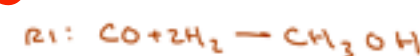
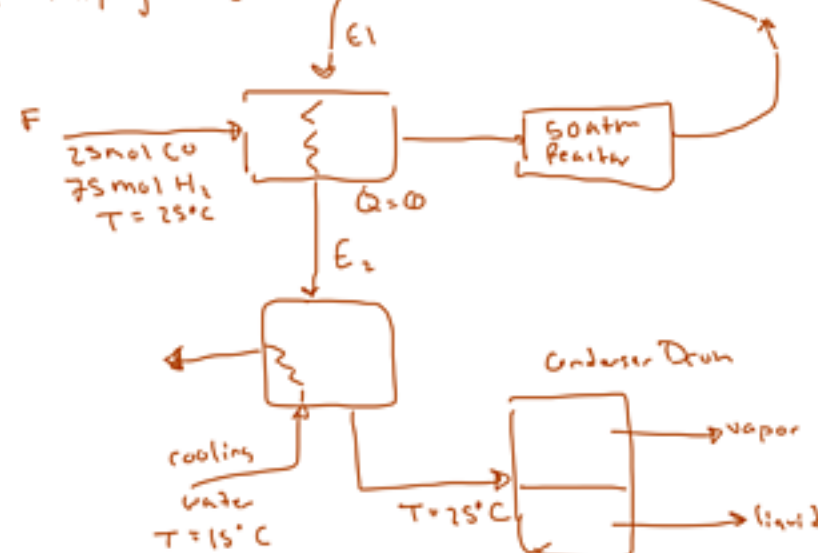
$$= -814.43 \text{ kJ}$$

$$\hat{Q} = \dot{Q}/n = -814.43 \text{ kJ/mol} = -814.43 \text{ kJ/mol}$$

c) explain

Only what reacts matters w/ heat transfer, and so the amt. of excess oxygen does not matter b/c it doesn't have heat formed. The inlet & outlet temps also have no influence, the oxygen reactant even affects the heat transfer even if it has heat capacity.

Q4. Murphy 6.68



Bal. on reactor

$$\text{conv. } 25(0.8) = 18.7 + R_{\text{P,CH}_4} \rightarrow R_{\text{P,CH}_4} = 1.3 \text{ mol/s CH}_4$$

$$\text{CO: } 25(0.2) = 25 - \dot{J}_1 - \dot{J}_2 \rightarrow 5 \text{ mol/s CO}$$

$$\text{H}_2: R_{\text{P,H}_2} = 72 - 2\dot{J}_1 - 3\dot{J}_2 \rightarrow 33.7 \text{ mol/s H}_2$$

$$\text{CH}_3\text{OH: } 18.7 = 0 + \dot{J}_1 \rightarrow \dot{J}_1 = 18.7 \text{ mol/s CH}_3\text{OH}$$

$$\text{CH}_4: 1.3 = 0 + \dot{J}_2 \rightarrow \dot{J}_2 = 1.3 \text{ mol/s CH}_4$$

$$\text{H}_2\text{O: } R_{\text{P,H}_2\text{O}} = 0 + \dot{J}_2 \rightarrow 1.3 \text{ mol/s H}_2\text{O}$$

$$\Delta H = \dot{Q} = \sum \dot{J}_i \Delta \hat{H}_i + \sum_{\text{out}} \dot{n}_i \hat{H}_{\text{out}} - \sum_{\text{in}} \dot{n}_i \hat{H}_{\text{in}}$$

$$\text{R1: } [(1 \text{ mol})(-200.94 \text{ kJ/mol}) - (1 \text{ mol})(-110.53 \text{ kJ/mol}) + (2 \text{ mol})(0)] = -90.41 \text{ kJ}$$

$$\text{R2: } [(1 \text{ mol})(-74.521) + (1 \text{ mol})(-241.83)] - (1 \text{ mol})(-110.53) + (3 \text{ mol})(0) = -205.82 \text{ kJ}$$

$$\Delta H = (18.7)(-90.41) + (1.3)(-205.82) = -1959.23 \text{ kJ}$$

Q5) $4 \text{ NH}_3(\text{g}) + 5 \text{ O}_2(\text{g}) \rightarrow 4 \text{ NO}(\text{g}) + 6 \text{ H}_2\text{O}(\text{g})$ Feed stream... 4 mol/s NH₃6 mol/s O₂

$$/ 2657 \text{ kJ} \quad 3626 \text{ K}$$

The Shomate eq. can be used down to 298 K because the Shomate eq. coeffs. are only fitted for avg/standard thermo data, & so the thermo data in Murphy may be the same/similar at the base of 298 K, which is room temp. The plot/graphs may have limits to their similarities at certain point, however, due to this difference in data.

no more time